

# Sequences (S)

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June 25, 2026

Here are some powerful techniques in solving questions with sequences.

## 1 Linear recurrences

### 1.1 Linear Homogeneous Recurrences

**Example 1:**  $a_{n+2} = 3a_{n+1} - 2a_n$ , where  $a_0 = 0, a_1 = 1$ .

We start with the *ansatz*  $a_n = x^n$ . This gets us

$$x^{n+2} = 3x^{n+1} - 2x^n$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$

Hence we conclude that  $a_n = 2^n$  and  $a_n = 1^n$  are solutions to the recurrence relation. Notice that any linear combination of these two solutions is also a solution to the recurrence. So the general solution to the recurrence is  $a_n = A2^n + B1^n$  for some constants  $A$  and  $B$ . Since  $a_0 = A + B = 0$  and  $a_1 = 2A + B = 1$ , we conclude that  $A = 1, B = -1$ . Hence, the solution to this recurrence is  $a_n = 2^n - 1$ .

**Example 2:**  $a_{n+2} = 4a_{n+1} - 4a_n$ , where  $a_0 = 1, a_1 = 6$ .

Using the same *ansatz*, we get the equation  $(x - 2)^2 = 0$ . Notice that we now have a root with multiplicity. So, alongside the usual solution of  $a_n = 2^n$ , we also have the solution  $a_n = n2^n$  (If it was a triple root, then we have the solution  $a_n = n^22^n$ , etc (check it!)). As before, the general solution is  $a_n = (A + Bn)2^n$  for some constants  $A, B$ , where we deduce  $A = 1$  and  $B = 2$  via the initial conditions. Thus, the solution is  $a_n = (1 + 2n)2^n$ .

**Exercises:** Find the general term.

1.  $a_{n+3} = 6a_{n+2} - 11a_{n+1} + 6a_n$ , where  $(a_0, a_1, a_2) = (-1, 0, 4)$ .
2.  $a_{n+3} = a_{n+2} + a_{n+1} - a_n$ , where  $(a_0, a_1, a_2) = (2, 2, 6)$ .
3.  $F_{n+2} = F_{n+1} + F_n$ , where  $(F_0, F_1) = (0, 1)$ .

**Problems:**

1. Find the general term of the fibonacci sequence  $F_{n+2} = F_{n+1} + F_n, F_0 = 0, F_1 = 1$ . Hence show that  $F_{n+1}/F_n \rightarrow \phi$ , where  $\phi = \frac{1+\sqrt{5}}{2}$  is the golden ratio.
2. Show that the general term of  $a_{n+2} = a_{n+1} - a_n$ , where  $(a_0, a_1) = (0, 1)$  is  $a_n = \frac{2}{\sqrt{3}} \sin\left(\frac{n\pi}{3}\right)$ .

3. Let  $\{a_n\}$  be a positive sequence such that  $a_{n+2} = -a_{n+1} + 2a_n$ . Show that  $a_n$  must be a constant sequence.
4. Let  $\{a_n\}$  be an integer sequence such that  $2a_{n+2} = 5a_{n+1} - 3a_n$ . Show that  $a_n$  must be a constant sequence.
5. Let  $\{a_n\}$  be a real valued sequence such that  $a_{n+2} - ka_{n+1} + a_n = 0$ , for some real number  $|k| < 2$ . Prove that  $a_n$  is bounded.
6. Find all functions  $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$  such that  $f(f(x)) = -4f(x) + 5x$ .
7. Find all functions  $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$  such that  $f(x) > x$  and  $f(f(x) - x) = 2x$ .
8. (ISL 2010 A5) Find all functions  $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$  such that  $f(f(x)^2y) = x^3f(xy)$ .
9. (China 2017) The sequences  $\{u_n\}$  and  $\{v_n\}$  are defined by  $u_{n+2} = 2u_{n+1} - 3u_n$ , where  $(u_0, u_1) = (1, 1)$  and  $v_{n+3} = v_{n+2} - 3v_{n+1} + 27v_n$ , where  $(v_0, v_1, v_2) = (a, b, c)$ . There exists a positive integer  $N$  such that for any  $n > N$ ,  $u_n$  divides  $v_n$ . Prove that  $3a = 2b + c$ .
10. (SMMC sample) Consider a circle with circumference  $\frac{1+\sqrt{5}}{2}$  and mark a point  $P_0$  on it. Going clockwise, you mark the points  $P_1, P_2, \dots$  for every unit arc length. Suppose you terminate at step  $n$ , after you mark the point  $P_n$ . Show that if  $P_i, P_j$  are adjacent on the circle, then  $|i - j|$  is a Fibonacci number.

## 1.2 Linear Inhomogeneous Recurrences

These are sequences with linear  $a_n$  terms, with along with some functions of  $n$ . The homogeneous version of the recurrence gives the general part of the solution, while you need a particular solution to take care of the inhomogeneous part (this is where you need to be smart).

**Example 1:**  $a_{n+1} = 3a_n - 2$ , where  $a_0 = 0$ .

The homogeneous version,  $a_{n+1} = 3a_n$  has an easy general solution  $a_n = A3^n$ . We can guess a particular solution to the original equation, which is  $a_n \equiv 1$ . Thus, the general solution is  $a_n = A3^n + 1$ , where we tweak  $A = -1$  to match the initial condition. Thus, the solution is  $a_n = 1 - 3^n$ .

The methodical way to do this is to rearrange the original equation to  $a_{n+1} - 1 = 3(a_n - 1)$ , and define  $b_n = a_n - 1$ . Then the  $b_n$  recurrence is  $b_{n+1} = 3b_n$ , etc. Sometimes guessing is not that easy, so these substitutions are useful.

**Example 2:**  $a_{n+1} - 5a_n = 5^{n+1}$ , where  $a_0 = 0$

Divide both sides with  $5^{n+1}$  to get  $\frac{a_{n+1}}{5^{n+1}} - \frac{a_n}{5^n} = 1$ . Define  $b_n = \frac{a_n}{5^n}$ , then we get  $b_{n+1} - b_n = 1$ , with  $b_0 = \frac{a_0}{5^0} = 0$ . This yields  $b_n = n$ , so we conclude  $a_n = n5^n$ .

**Problems:**

1.  $a_{n+1} - 5a_n = (2n + 1)5^{n+1}$ , where  $a_0 = 0$ .
2. (AMO 2015) Define the sequence  $\{a_n\}$  such that  $a_{n+2} = 2a_{n+1} - a_n + 2$  with  $(a_0, a_1) = (4, 7)$ . Show that  $a_n a_{n+1}$  is always a term of the sequence.
3. (102 PiC) How many subsets of  $\{1, 2, 3, \dots, 2020\}$  has the property that the sum of its elements is a multiple of 5?

### 1.3 First Order Multi-variable Homogeneous Recurrences

Sometimes you have more than one sequence working in tandem to define each other.

**Example:** Find the number of binary sequences of length  $n$  such that there does not exist consecutive zeros.

Let  $a_n, b_n$  be the number of binary sequences of length  $n$  ending with 0 and 1 respectively. As there cannot be consecutive zeros, we have  $a_{n+1} = b_n$  (every sequence counted in  $a_{n+1}$  is a 0 subtended to a  $b_n$  sequence) and  $b_{n+1} = a_n + b_n$  (every sequence counted in  $b_{n+1}$  is 1 subtended to a  $a_n$  or  $b_n$  sequence).

Since we're only interested in the sum  $x_n := a_n + b_n$ , let us inspect this instead.

$$x_n = a_n + b_n = b_{n-1} + b_n = (a_{n-2} + b_{n-2}) + (a_{n-1} + b_{n-1}) = x_{n-2} + x_{n-1}$$

Hence the recurrence is the Fibonacci recurrence.

What if the double-recurrence isn't nice enough so that such manipulations are not immediately obvious?

Our double recurrence can be expressed as  $\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}$ . Let  $\mathbf{x}_n := \begin{pmatrix} a_n \\ b_n \end{pmatrix}$  and  $A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ . Then  $\mathbf{x}_{n+1} = A\mathbf{x}_n$ . Thus, our general form becomes  $\mathbf{x}_n = A^n \mathbf{x}_0$ . The real task is figuring out how to find  $A^n$  in a reasonable amount of time. You can *diagonalise* the matrix such that  $A = PDP^{-1}$  for some diagonal matrix  $D$  (which means we can find  $D^n$  easily), so we get  $A^n = PD^nP^{-1}$ , which is actually solvable (details not included for sanity; look up how to diagonalise a matrix).

### 1.4 Pell equations

In number theory, this is a Pell equation:  $x^2 - dy^2 = 1$  for a non-square number  $d$ .

**Example:**  $x^2 - 2y^2 = 1$ .

We need an initial solution. There is no method to find this, so good luck with your guesses. In this case,  $(x, y) = (3, 2)$  is a solution. Let  $(x_n, y_n)$  be a solution to the equation. Then we do a weird factorisation.

$$\begin{aligned} (x_n - y_n\sqrt{2})(x_n + y_n\sqrt{2}) &= 1 \\ (3 - 2\sqrt{2})(3 + 2\sqrt{2}) &= 1 \end{aligned}$$

Multiplying the terms at their respective locations,

$$((3x_n + 4y_n) - (2x_n + 3y_n)\sqrt{2})((3x_n + 4y_n) + (2x_n + 3y_n)\sqrt{2}) = 1$$

So we see that  $(x_{n+1}, y_{n+1}) = (3x_n + 4y_n, 2x_n + 3y_n)$  is a new solution to this Pell equation.

Before you start to write matrices, look a bit closer on how we generated the solutions.  $x_n \pm y_n\sqrt{2} = (3 \pm 2\sqrt{2})^n$ .

$$x_n = \frac{(3 + 2\sqrt{2})^n + (3 - 2\sqrt{2})^n}{2} \quad y_n = \frac{(3 + 2\sqrt{2})^n - (3 - 2\sqrt{2})^n}{2\sqrt{2}}$$

**Problems:**

- (AMO 2020) Sequences  $\{a_n\}$  and  $\{b_n\}$  satisfy  $a_1 = b_1 = 1$  and  $a_{n+1} = \frac{a_n+2}{a_n+1}$  and  $b_{n+1} = \frac{b_n}{2} + \frac{1}{b_n}$ . Show that  $b_{n+1} = a_{2^n}$ .
- (Crux)  $x_{n+2} = x_{n+1}\sqrt{x_n^2 + 1} + x_n\sqrt{x_{n+1}^2 + 1}$  satisfies  $x_0 = x_1 = 1$ . Find its general form.
- (Jongmin) Positive integers  $x, y$  satisfy  $x^2 = (3y + 4)(y + 1)$ . Find all solutions.

## 2 Telescoping

We want to turn the summation or product into the following forms to simplify nicely.

$$\sum_{k=1}^n a_k - a_{k-1} = a_n - a_0$$

$$\prod_{k=1}^n \frac{a_k}{a_{k-1}} = \frac{a_n}{a_0}$$

### Exercises:

1. Simplify  $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{9999 \times 10000}$
2. Simplify  $\sum_{k=0}^{2017} \frac{k}{(k+1)!}$
3. Simplify  $\sum_{k=0}^{2017} \binom{k+8}{k}$
4. Simplify  $\prod_{k=1}^{2017} \frac{k^2 + 5k + 4}{k^2 + 5k + 6}$
5. Prove that  $1.5 < \sum_{n=1}^{\infty} \frac{1}{n^2} < 2$

### Problems:

1. (Balkan 2016) Find all injective functions  $f: \mathbb{R} \rightarrow \mathbb{R}$  such that for  $x \in \mathbb{R}$  and  $n \in \mathbb{Z}^+$ ,

$$\left| \sum_{i=1}^n i (f(x+i+1) - f(f(x+i))) \right| < 2016$$

2. (ISL 2015) Suppose that a sequence  $a_1, a_2, \dots$  of positive real numbers satisfies

$$a_{k+1} \geq \frac{ka_k}{a_k^2 + (k-1)}$$

for every positive integer  $k$ . Prove that  $a_1 + a_2 + \dots + a_n \geq n$  for every  $n \geq 2$ .

3. (USAMTS) Determine the value of

$$S = \sum_{k=1}^{2016} \sqrt{1 + \frac{1}{k^2} + \frac{1}{(k+1)^2}}$$

4. (CMC 2020) Let  $a_1, a_2, a_3 \dots$  be the sequence of positive real numbers such that for each  $n \in \mathbb{Z}^+$ ,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt{\frac{a_1^2 + a_2^2 + \dots + a_{n+1}^2}{n+1}}$$

Show that  $\{a_n\}$  is a constant sequence.

### 3 Convergence

**Completeness axiom of real numbers:** Every subset  $S \subseteq \mathbb{R}$  with an upper bound has a *least upper bound* in  $\mathbb{R}$ , also known as the supremum  $s$ . Note that this axiom does not necessitate that  $s \in S$ .

**Exercises:**

1. Let  $A = \{n \in \mathbb{Z}^+ \mid 1 - 2^{-n}\}$ . What is the supremum  $s$  of  $A$ ? Is  $s$  an element of  $A$ ?
2. Show that every subset  $S \subseteq \mathbb{R}$  with a lower bound has a *greatest lower bound* in  $\mathbb{R}$ , also known as the infimum.
3. Show that this property does not hold for  $\mathbb{Q}$ . Can you think of an example of a bounded subset of  $\mathbb{Q}$  that does not have a supremum in  $\mathbb{Q}$ ?
4. Show that every monotonically increasing sequence with an upper bound converges to a finite limit.
5. Show that every monotonically decreasing sequence with a lower bound converges to a finite limit.
6. Let  $a_0 = 2$  and  $a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}$ . Show that  $a_n$  converges. What is the limit?

**Problems:**

1. Let  $a_n = (-2)^n \sin n\theta$  for some fixed  $\theta \neq 0$ . Show that  $a_n$  is unbounded.
2. Let  $0 \leq f(x) < 1$  denote the fractional part of  $x$ ; i. e.  $f(1.6) = 0.6$ ,  $f(-4.2) = 0.8$ . Let  $\alpha \in \mathbb{R}$ . Show that, for any  $\epsilon > 0$ , there exists  $N \in \mathbb{Z}^+$  such that  $0 \leq f(N\alpha) < \epsilon$ .
3. (Jongmin) Consider a sequence given by  $a_{n+1} = \sqrt{2^{a_n}}$ ,  $a_0 = 1$ .
  - (a) Show that  $\{a_n\}$  converges. What is the limit? Why is the limit not 4?
  - (b) For which values of  $a_0$  does  $\{a_n\}$  converge? What number does it converge to for each value of  $a_0$ ?
4. Let  $\{a_n\}$  be decreasing positive sequence converging to zero. Define the sequence  $\{S_n\}$  by  $S_n = \sum_{k=0}^n (-1)^k a_k$ . Show that  $\{S_n\}$  converges.
5. (AMO 1989)  $u_n$  satisfies  $u_1 = 1$ , and  $u_1 + u_2 + \dots + u_n = \frac{1}{u_{n+1}}$ . Determine whether there exist  $m$  such that

$$\sum_{i=1}^m u_i \geq 1989$$

6. (SMMC 2017) Let  $a_1, a_2, a_3 \dots$  be the sequence of real numbers defined by  $a_1 = 1$  and for  $m \geq 2$

$$a_m = \frac{1}{a_1^2 + a_2^2 + \dots + a_{m-1}^2}$$

Show whether there exist  $N$  such that

$$a_1 + a_2 + \dots + a_N > 2017^{2017}$$

## 4 Generating functions

Given a sequence  $\{a_n\}$ , we consider a formal power series  $f(x) = a_0 + a_1x + a_2x^2 + \dots$ . We call this the generating function corresponding to the sequence  $\{a_n\}$ . This is useful as we will see very soon.

### 4.1 How to get generating functions from sequences

**Example 1:** What is the generating function of  $a_n = 2^n$ ?

Indeed the generating function is  $f(x) = 1 + 2x + 4x^2 + \dots$ . For generating function purposes, we don't really care about the technicalities of convergence. We just use the geometric series formula to simplify  $f(x) = \frac{1}{1-2x}$ .

**Example 2:** What is the generating function of  $a_n = n$ ?

Start with the generating function  $f(x) = \frac{1}{1-x} = 1 + x + x^2 + \dots$ . We can differentiate both sides to obtain

$$f'(x) = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \dots = \sum_{n=0}^{\infty} (n+1)x^n$$

However notice that our terms are off by one (we want  $\sum_{n=0}^{\infty} nx^n$ ). We can fix this easily by multiplying  $x$  to both sides. Thus the generating function is  $xf'(x) = \frac{x}{(1-x)^2}$ .

**Example 3:** What is the generating function of  $a_{n+2} = 3a_{n+1} - 2a_n$  with  $a_0 = 2$  and  $a_1 = 3$ ?

Let  $f(x) = a_0 + a_1x + a_2x^2 + \dots$  be the generating function. Then notice

$$\begin{aligned} f(x) &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots + a_{n+2}x^{n+2} + \dots \\ xf(x) &= a_0x + a_1x^2 + a_2x^3 + a_3x^4 + \dots + a_{n+1}x^{n+2} + \dots \\ x^2f(x) &= a_0x^2 + a_1x^3 + a_2x^4 + \dots + a_nx^{n+2} + \dots \end{aligned}$$

Thus according to the formula, most of the terms will cancel out in  $f(x) - 3xf(x) + 2x^2f(x)$ . Let's see what terms are left.

$$f(x) - 3xf(x) + 2x^2f(x) = a_0 + a_1x - 3a_0x = 2 - 3x$$

Thus the generating function is  $f(x) = \frac{2-3x}{1-3x+2x^2}$ . However, notice the following

$$\begin{aligned} f(x) &= \frac{2-3x}{1-3x+2x^2} \\ &= \frac{2-3x}{(1-x)(1-2x)} \\ &= \frac{1}{1-x} + \frac{1}{1-2x} \\ &= (1 + x + x^2 + \dots) + (1 + 2x + 4x^2 + \dots) \\ &= \sum_{n=0}^{\infty} (1 + 2^n)x^n \end{aligned}$$

Which is consistent with our understanding of this sequence, because the general term is indeed  $a_n = 1 + 2^n$ .

**Exercises:**

1. What is the generating function for  $a_n = n^2$ ?
2. What is the generating function for  $a_n = n2^n$ ?
3. What is the generating function for the Fibonacci sequence given by  $F_{n+2} = F_{n+1} + F_n$  and  $F_0 = F_1 = 1$ ?

## 4.2 How to get sequences back from generating functions

**Example 1:** What is the sequence defined by the generating function  $f(x) = e^x$ ?

We use the Taylor series. The  $n!a_n$  would be the constant term after differentiating  $n$  times.

$$a_n = \frac{1}{n!} \left[ \frac{d^n}{dx^n} f(x) \right]_{x=0} = \frac{1}{n!} [e^x]_{x=0} = \frac{1}{n!}$$

Thus  $f(x)$  defines the sequences  $a_n = \frac{1}{n!}$ .

**Exercises:**

1. What is the sequence given by  $f(x) = \sqrt{1+x}$ ?
2. What is the sequence given by  $f(x) = \ln(x+1)$ ?

## 4.3 How to use generating functions to count things

Consider  $a_n$  as the number of ways to choose  $n$  things satisfying some condition, with power series  $f(x) = a_0 + a_1x + a_2x^2 + \dots$ . Similarly assume we have another sequence  $\{b_n\}$  and the power series  $g(x) = b_0 + b_1x + b_2x^2 + \dots$ , and say  $b_n$  is the number of ways to choose  $n$  things satisfying another condition. One might ask, how many ways can we choose  $n$  things that satisfies both conditions? We can choose  $k$  things satisfying the first condition, and  $n-k$  things satisfying the second condition, and add it all up for  $k=0$  to  $k=n$ . If we call this  $c_n$ , then we have

$$c_n = a_0b_n + a_1b_{n-1} + \dots + a_{n-1}b_1 + a_nb_0$$

Coincidentally, this is exactly what happens to polynomial coefficients when you multiply them (wow!). Thus our new generating function is  $h(x) = c_0 + c_1x + c_2x^2 + \dots = f(x)g(x)$ .

**Example 1:** You have 1 apples, 2 bananas, 3 carrots. How many ways can you choose 4 things?

The gf for apples is  $a(x) = 1+x$ , the gf for bananas is  $b(x) = 1+x+x^2$  and the gf for carrots is  $c(x) = 1+x+x^2+x^3$ . The gf for the whole thing is  $a(x)b(x)c(x) = (1+x)(1+x+x^2)(1+x+x^2+x^3) = x^6 + 3x^5 + 5x^4 + 6x^3 + 5x^2 + 3x + 1$ . Since the coefficient for  $x^4$  is 5, the answer is 5.

**Example 2:** How many ways can you choose  $k$  people out of a class of  $n$  people?

The gf for a single person is  $1+x$ , since there's one way to choose zero people out of this one person, and one way to choose one person out of this person (weird phrasing I know). Now, the gf for  $n$  people is  $(1+x)(1+x)\dots(1+x) = (1+x)^n$ . We all know that the coefficient of the  $n$ -th term is  $\binom{n}{k}$  which is indeed the answer.

**Example 3:** At a supermarket, there are  $n$  different items. How many ways can you choose  $k$  things allowing repetition?

The gf for a single item is  $\frac{1}{1-x} = 1 + x + x^2 + \dots$ , since there's one way to choose zero things, one way to choose one thing, one way to choose two things, etc. Now, the gf for  $n$  items is  $\frac{1}{(1-x)^n}$ . We can obtain the sequence back

from the generating function by using the following observation

$$\begin{aligned}(n-1)! \frac{1}{(1-x)^n} &= \frac{d^{n-1}}{dx^{n-1}} \frac{1}{1-x} \\(n-1)! \frac{1}{(1-x)^n} &= \frac{d^{n-1}}{dx^{n-1}} \sum_{k=0}^{\infty} x^k \\(n-1)! \frac{1}{(1-x)^n} &= \sum_{k=0}^{\infty} (n+k-1)(n+k-2) \cdots (k+1) x^k \\ \frac{1}{(1-x)^n} &= \sum_{k=0}^{\infty} \frac{(n+k-1)(n+k-2) \cdots (k+1)}{(n-1)!} x^k\end{aligned}$$

Thus we retrieve the formula for combination with repetition  $\binom{n+k-1}{k}$ .

**Example 4:** Let the Fibonacci sequence be given by  $F_{n+2} = F_{n+1} + F_n$  and  $F_0 = F_1 = 1$ . What is  $\sum_{n=0}^{\infty} F_n 2^{-n}$ ? Sometimes when we know the power series converges, we can calculate interesting sums. We can find that the generating function is

$$F(x) = F_0 + F_1 x + F_2 x^2 + \cdots = \frac{1}{1-x-x^2}$$

Subbing in  $x = \frac{1}{2}$ , we see that  $\sum_{n=0}^{\infty} F_n 2^{-n} = F(\frac{1}{2}) = 4$ .

#### Exercises:

1. You have apples, bananas, carrots, and durians. You have to choose a multiple of 5 apples, even number of bananas, at most 1 carrot and at most 4 durians. How many ways can you choose 2020 things?

#### Problems:

1. (102 PiC) How many subsets of  $\{1, 2, 3, \dots, 2020\}$  has the property that the sum of its elements is a multiple of 5?
2. Consider an  $n \times n$  grid. Let  $c_n$  be the number of shortest paths along the gridlines from the bottom left corner to the top right corner. Find the general form for  $c_n$ . What is  $\sum_{n=0}^{\infty} \frac{c_n}{4^n}$ ?
3. An alternating permutation is a permutation of  $\{1, 2, \dots, n\}$  such that  $a_1 < a_2 > a_3 < a_4 > a_5 \cdots$ . How many alternating permutations are there?
4. (SMMC 2018) For each positive integer  $n$ , consider a cinema with  $n$  seats in a row, numbered left to right from 1 up to  $n$ . There is a cup holder between any two adjacent seats and there is a cup holder on the right of seat  $n$ . So seat 1 is next to one cup holder, while every other seat is next to two cup holders. There are  $n$  people, each holding a drink, waiting in line to sit down. In turn, each person chooses an available seat uniformly at random and does the following:
  - (a) If they sit next to two empty cup holders, then they choose a cup holder at random and put their drink in it
  - (b) If they sit next to one empty cup holder, then they place their drink in that cup holder.
  - (c) Otherwise, they get angry.

Let  $p_n$  be the probability that no one gets angry. What is  $p_1 + p_2 + p_3 + \cdots$ ?